

Unsupervised Deep Learning Can Identify Protein Functional Groups from Unaligned Sequences

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Accepted: 13 May 2023

Abstract

Interpreting protein function from sequence data is a fundamental goal of bioinformatics. However, our current understanding of protein diversity is bottlenecked by the fact that most proteins have only been functionally validated in model organisms, limiting our understanding of how function varies with gene sequence diversity. Thus, accuracy of inferences in clades without model representatives is questionable. Unsupervised learning may help to ameliorate this bias by identifying highly complex patterns and structure from large data sets without external labels. Here, we present DeepSeqProt, an unsupervised deep learning program for exploring large protein sequence data sets. DeepSeqProt is a clustering tool capable of distinguishing between broad classes of proteins while learning local and global structure of functional space. DeepSeqProt is capable of learning salient biological features from unaligned, unannotated sequences. DeepSeqProt is more likely to capture complete protein families and statistically significant shared ontologies within proteomes than other clustering methods. We hope this framework will prove of use to researchers and provide a preliminary step in further developing unsupervised deep learning in molecular biology.

Key words: machine learning, protein annotation, bioinformatics.

Significance

In this manuscript, we report the results of a new unsupervised machine learning software, DeepSeqProt. Unsupervised methods offer several advantages which can help escape longstanding pitfalls and biases pervading computational molecular biology. DeepSeqProt learns from and processes unaligned protein sequences with the goal of clustering them into informative groups with regard to protein family and function, as well as distributing the clusters themselves in a lower dimension space. We discovered that unsupervised deep learning is capable of recognizing patterns shared among proteins of similar families and functional affinities, exceeding conventional sequence similarity-based clustering in some scenarios. DeepSeqProt has broad applications for computational molecular biology and may be especially useful for nonmodel organisms.

Introduction

As sequencing technology continues to improve, there is an ever-increasing need to adequately annotate and characterize novel protein sequences and their predicted functions. With thousands of new sequences being uploaded every day, predicting the function of every protein directly

with conventional experimental studies such as gene knockouts or assays is not possible. Thus, attempting to infer protein function automatically is necessary. Many such methods exist but fundamentally operate the same way: by matching the sequence of a protein with unknown function to a reference sequence of a protein with known function and then assuming that functions are the same. These

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matches are generally identified by sequence similarity, as inferred through methods such as the Blast algorithm (Altschul et al. 1990), closely related homology, hidden Markov models such as HMMER (Finn et al. 2011), or orthology using tree inference methods such as OrthoFinder (Emms and Kelly 2015, 2019). Of the 219,740,215 proteins listed in the UniProtKB database (Consortium 2018; UniProt Consortium 2021) (release 2021_03), only 0.77% had experimental evidence, compared with 31% with homology-based evidence. This reference-based approach is clearly a powerful tool to assign putative identity to unknown protein sequences. However, a consequence of this approach is that our assessment of protein function and diversity is restricted to only those sequences with sufficient pairwise site similarity to reference sequences which have been experimentally validated in model taxa. For example, in the Gene Ontology database (Ashburner et al. 2000; Gene Ontology Consortium 2021) (release 2021-07), 73% of annotations with experimental evidence belong to just five species (Carbon et al. 2009). Three of these five (human, rat, and mouse) belong to the same superorder. Reference-based approaches may cause researchers to miss or underestimate protein functional diversity, particularly in nonmodel taxa. For example, clades like cnidarians which are more phylogenetically distant from model species have larger “dark” regions of the proteome about which little is known (supplementary fig. S1, Supplementary Material online). This trend can create bias, leading researchers to conflate organismal complexity with research attention (Dunn et al. 2015; Dunn and Ryan 2015; David et al. 2019).

Failure to account for molecular diversity of nonmodel organisms will lead to a systemic underestimation of gene family size in these groups (Tassia et al. 2017), which can compromise functional studies such as those dependent on reporter constructs (Cavalieri and Spinelli 2014) or CRISPR-Cas9 (Connahs et al. 2019). Phylogenetically informed gene sampling is also necessary for assessing genotype-to-phenotype relationships, as assessing most phenotypic diversity is impossible with classical genetics (Smith et al. 2020). Finally, broad sampling is a prerequisite to address evolutionary questions that are comparative by nature. For example, a 2018 study drawing on data from across Bilateria identified multiple independent origins of nerve cords within the group, upending the previous consensus of a single origin based on just a few taxa (Martín-Durán et al. 2018). As nonmodel organisms become increasingly relevant in answering biological questions, there is a clear need to develop *ab initio* methods for characterizing novel protein sequences and their functions.

One possible avenue for exploring protein function without directly relying on model systems is through unsupervised deep learning (see table 1 for glossary of terms).

Deep learning refers to a class of machine learning methods characterized by multilayer neural networks designed to identify progressively complex patterns from large data sets (Zou et al. 2019). Unsupervised deep learning (in contrast to supervised learning) does not learn from externally provided examples or labels, but rather attempts to learn patterns inherent in data to construct a generalized model (Bengio et al. 2012). Several supervised deep learning methods are already being developed for protein classification which may outperform conventional techniques (Seo et al. 2018; Bileschi et al. 2019; Sureyya Rifaoglu et al. 2019; Feldbauer et al. 2020). However, as these supervised methods are reliant on high-confidence training examples, they are not able to escape the fundamental biases associated with reference-based methods. We are aware of one other unsupervised learning approach that has been developed for protein classification, with promising results (Rives et al. 2021).

One common implementation of unsupervised deep learning is through the use of variational autoencoders (Kingma and Welling 2014) (VAEs). VAEs have become increasingly popular in recent years (Doersch 2016) and have been developed in biology for problems such as predicting effects of mutations (Riesselman et al. 2018), visualizing population structure (Battey et al. 2021), and species delimitation (Derkarabetian et al. 2019). A VAE consists of two neural networks, an encoder and a decoder. The encoder q compresses input data x into a latent representation z from parameters θ : $q_{\theta}(z|x)$. The decoder p then samples the latent representation and attempts a reconstruction of data x' from parameters ϕ : $p_{\phi}(x'|z)$. The loss function can be approximated as $x' \approx p_{\phi}(q_{\theta}(x))$. The VAE tries to optimize θ and ϕ to produce the highest fidelity reconstruction of x after it has been mapped to the lower-dimensional z . Once trained, the learned distribution can then be sampled to generate its own data. For example, a VAE trained on images of objects (Dosovitskiy et al. 2017), animals (Razavi et al. 2019), or faces (Razavi et al. 2019) may be able to generate convincing original images. In biology, one recent study used a VAE to simulate realistic human genotypes (Montserrat et al. 2019).

For the purposes of this study, however, we are less interested in the generative power of these models than the latent representation itself. The VAE is forced to map input efficiently into a reduced latent representation with the expectation that it will learn key identifiable features from training data and distribute latent variables in a way that is meaningful to researchers (e.g., by placing proteins of the same family or with the same function close together). Such a method is still ultimately reliant on ground-truth experimental data from model systems to validate and interpret model output. However, unlike conventional annotation, the operation and performance of the model itself is reference-free. In this way, a VAE can be thought

Table 1

Glossary of Terms

Unsupervised learning	A type of machine learning where the algorithm learns to recognize patterns and structure in training data without explicit feedback or manual labels. Unsupervised learning algorithms are able to identify relationships and/or classifications from the data without being provided with examples or labels.
Neural network	A class of machine learning algorithms characterized by layers of nodes (or “neurons”) connected by multipliers (weights). Nodes are combined with addends (biases) and passed to an activation function, whose output is then passed to the next node (if applicable). The network “learns” by adjusting its weights and biases to optimize its loss function through a process called backpropagation. Neural networks consist of an input layer, an output layer, and one or more hidden (latent) layers (supplementary fig. S2A, Supplementary Material online).
Deep learning	A neural network with multiple hidden layers (generally three or more) (supplementary fig. S2B, Supplementary Material online).
Feedforward network	An acyclic neural network (in contrast to recurrent neural networks).
Variational autoencoder (VAE)	A type of neural network used to represent and generate complex data in a low-dimensional latent space. A neural network (the encoder) maps input data into embeddings which are then mapped back into the original input space by a decoder.
Vector quantized variational autoencoder (VQ-VAE)	A variant of variational autoencoders that uses a discrete distribution to map embeddings into latent space, rather than a gaussian one.
Embedding	A learned numeric vector representing a lower dimensionality reduction of a discrete variable. Useful for quantifying, classifying, or visualizing higher-order concepts.
Loss function	A function used to assess the performance of a machine learning model, generally by comparing the expected and observed output. The model “learns” by optimizing this function to reduce loss (error).
Fully connected layer	A layer in which all nodes are connected to all nodes in the subsequent layer.
Convolutional layer	A layer in which only nearby nodes share connections to the subsequent layer, through the use of a sliding window called a kernel. Convolutional layers are used to identify local features in the data and are especially popular in image recognition. Convolutional layers are defined by several parameters (supplementary fig. S2C, Supplementary Material online): Kernel size: The size of the sliding window. Stride: How many “steps” the sliding window takes between each kernel. Dilation: A factor used to expand the kernel by inserting empty spaces, making it more “porous” (and thereby skipping certain inputs).
Transposed convolutional layer	A convolutional layer which inserts zero values into the input in order to perform upsampling, resulting in a larger data output size (as opposed to the smaller output produced by standard convolutional layers).

of as an analog to principal component analysis (PCA), although VAEs allow for nonlinear relationships which may enable them to capture greater variation in data (Kramer 1991; Riesselman et al. 2018; Battey et al. 2021).

Most VAEs use a multivariate normal distribution, but, in theory, almost any type of distribution should work. One variant that has recently shown promising results, known as a vector quantized variational autoencoder (VQ-VAE), uses a discrete distribution (van den Oord et al. 2017). The intuition behind such an approach posits that a discrete distribution may be more appropriate for categorical data. For example, with image data sets like CIFAR-10, each discrete latent variable could ideally represent one of the image classes (dog, frog, horse, etc.).

To explore applications unsupervised deep learning may have for molecular biology, we created DeepSeqProt, a VQ-VAE which accepts unaligned protein sequence data as input. We wrote DeepSeqProt in part to explore how the latent embeddings produced by VQ-VAEs reflect protein functional diversity as it is currently understood (e.g., canonical protein families). We are further interested in

the utility unsupervised deep learning may have for protein clustering and annotation generally.

Results and Discussion

Both DeepSeqProt models (default and 2D convolutional) were trained on the 1,612 eukaryotic reference proteomes currently available in the UniProt database (release 2021_03). Training was performed on a single NVIDIA T4 Tensor Core GPU. The default model converged after ~4 h, and the 2D convolutional model converged after ~1.5. Once trained, models processed training proteomes on a standard Intel Xeon Gold 6248R CPU. Testing the model took ~2 min for all three proteomes for both models, or ~2–3 s per every 1,000 sequences.

Clustering

Clusters produced from DeepSeqProt provide a distinct view of protein organization compared with conventional clustering methods based on pairwise similarity between

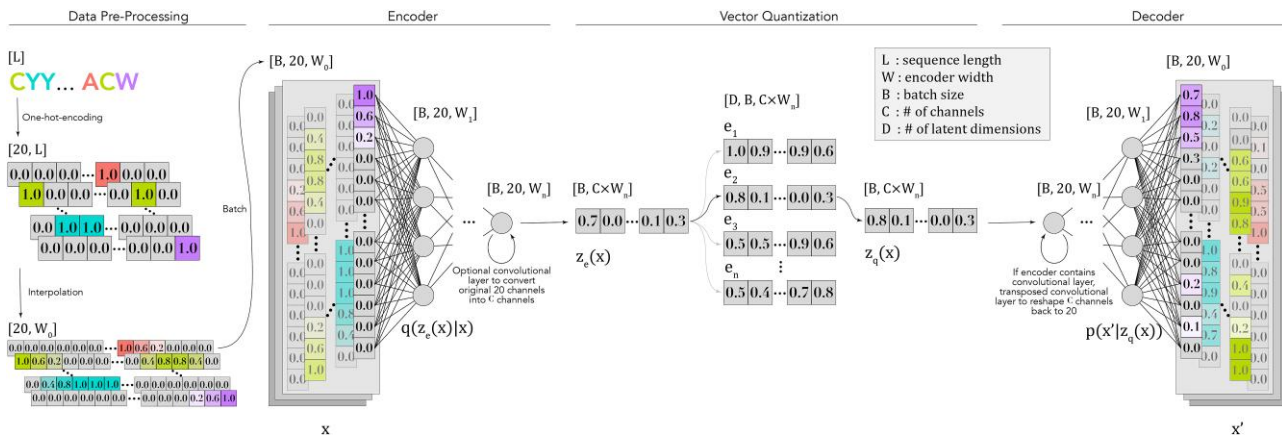


FIG. 1.—Overview of DeepSeqProt. Bracketed values represent the data shape at each step.

sites (fig. 2). DeepSeqProt had the highest family completeness statistic across taxa, from 0.68 in *Mus musculus* to 0.76 in *Saccharomyces cerevisiae*, offering an average 0.31 increase over conventional clustering. In contrast, AMI values for DeepSeqProt were consistently lower compared with other methods, with an average decrease of 0.26. These results indicate that although DeepSeqProt clusters were more inclusive to members of the same protein family, they were less likely to resolve protein families as homogeneous sets. With regard to Gene Ontology (GO) accuracy, DeepSeqProt outperformed other methods, with an average increase of 0.35. This is likely because conventional clustering methods relying on pairwise site similarity can produce very granular results. Clustering methods used for our benchmarks produced many, smaller clusters that contained too few representatives to provide statistical power to enrichment analysis. This effect is especially apparent in the *S. cerevisiae* proteome due to its smaller size (fig. 2), where as few as 2% of clusters had significant enrichment of any ontology.

DeepSeqProt consistently outperformed all other benchmarks with regard to protein family completeness and gene ontology accuracy but underperformed on adjusted mutual information (AMI). Overall, these results indicate that DeepSeqProt generates fewer, larger clusters made up of multiple protein families that generally share functional similarity, but do not adequately reflect individual families. One strength of neural networks for analyzing biological sequences is their ability to analyze latent factors between sites across the entire length of the sequence, compared with sitewise factors used by conventional clustering (Riesselman et al. 2018). Therefore, DeepSeqProt is, perhaps unsurprisingly, more “lumper” than “splitter” compared with conventional clustering, as single site differences are less likely to dissuade the model from using the same cluster for similar proteins. Despite the lack of specificity, groups assigned by DeepSeqProt clusters may still

be of use for researchers interested in capturing the entire diversity of a protein family with less regard to how many other similar proteins may be included as well or for capturing broad functional classes. For example, DeepSeqProt may serve as a useful precursor to homology inference or tree inference steps in programs like OrthoFinder (Emms and Kelly 2019). DeepSeqProt will also likely capture more diverse homologs from distantly related or underrepresented taxa than conventional clustering, as it is less dissuaded by single site sequence dissimilarity.

Visualization

Embeddings trained in a two- or three-dimensional latent space offer a way to visualize protein sets along axes that contain biologically relevant information. In this way, DeepSeqProt performs similarly to other dimensionality reduction techniques such as UMAP (McInnes et al. 2018) or t-SNE (Van der Maaten and Hinton 2008) (though a recent study suggests visualizations produced by such methods are more arbitrary than previously suspected (Chari et al. 2021)). We find that DeepSeqProt embeddings preserve the general local and global structure of proteome functional diversity (fig. 3). For example, embeddings whose clusters are most associated with DNA-templated transcription are located closely together within latent space while also coinciding with other clusters associated with RNA polymerase II transcription. This “transcription space” can be contrasted with “signaling space” around quadrant IV of figure 3 where cell membrane signaling associated clusters are found, home to families like G proteins or defensin-like proteins in the case of *Arabidopsis*. This type of structure is apparent across all three aspects of gene ontology and shared across eukaryotic kingdoms. These visualizations demonstrate the generality and versatility of DeepSeqProt in “mapping” protein sequence data in functional space and may be a useful tool in categorizing

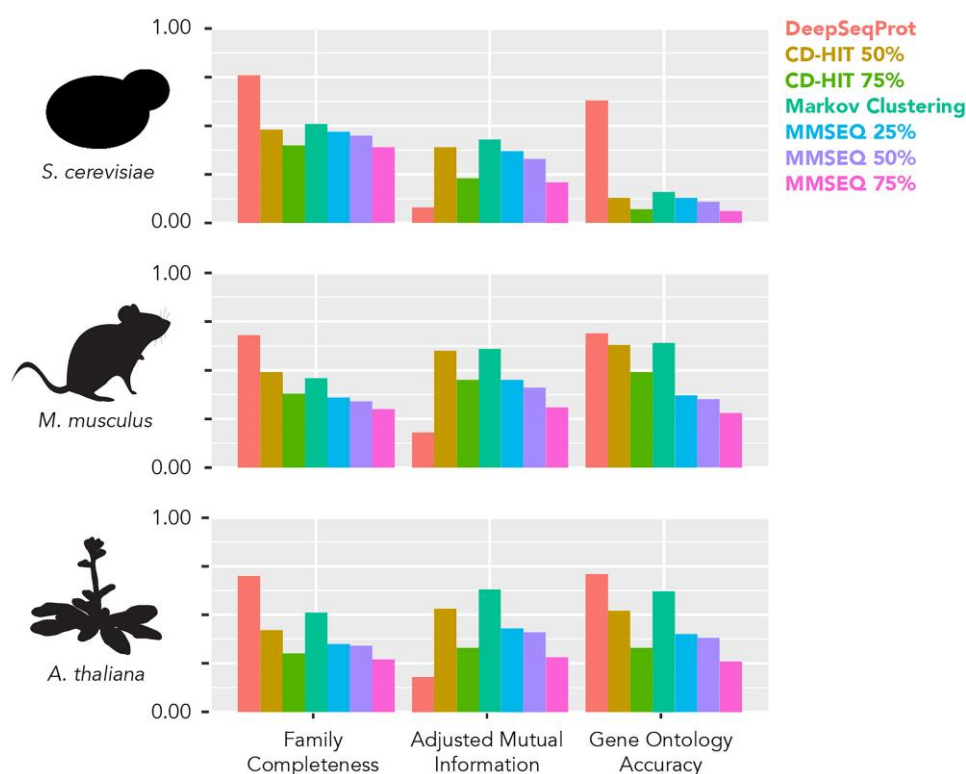


FIG. 2.—Benchmarking results.

proteins from diverse taxonomic groups distantly related to model organisms.

What Can Unsupervised Learning Teach Us about Molecular Diversity?

Traditional methods of assessing protein function are reliant on connecting new sequences to a small number of high-confidence references. Although this approach is powerful, it is also necessarily limited. Such methods will at best underestimate the full breadth of molecular diversity and at worst create systematic bias, leading to a warped perception of biology (Dunn and Ryan 2015; Tassia et al. 2021). Unsupervised learning provides an alternative approach. Information is not tied to specific references but instead can be produced from the data itself in aggregate. Given enough data an adequately specified model will learn complex features and relationships of novel proteins on its own, providing a reference-agnostic picture of protein functional diversity. One potential advantage to this approach is reduced bias; training data sets are limited only by the availability of sequence data and can otherwise be as taxonomically balanced as the user wants, rather than relying on similarity to a handful of model organisms. The tradeoff for this reduced bias is a lack of specificity. For example, DeepSeqProt is unable to report with confidence whether a protein belongs to a particular family, and

reference information is still necessary to interpret its results. However, reference data do not contribute to the operation of the model itself.

Unsupervised learning may also serve as a way to benchmark traditional methods in order to “keep them honest.” For example, if proteins from a nonmodel organism are routinely associated with the same clusters and regions of latent space as known cell-signaling proteins, but lack any cell-signaling domains themselves, it may indicate a lack of information in reference databases or an undiscovered protein family. This is especially true for lineages that are more distantly related from model organisms. For example, DeepSeqProt may be used to identify putative, lineage-specific protein families in clades without model representatives such as annelids, mollusks, or many fungal clades (supplementary fig. S1, Supplementary Material online).

We predict that both unsupervised approaches and traditional reference-based methods will be important to the future of protein annotation. As we demonstrate here, autoencoders are more likely to generate fewer, larger groups compared with traditional clustering. This offers a “greedy” approach for identifying diverse proteins which largely fall under the same functional classes. These groups can then be examined themselves using more conventional approaches to assign specific annotations. Although this study focuses on high-confidence reference proteomes for validation purposes, DeepSeqProt can operate on

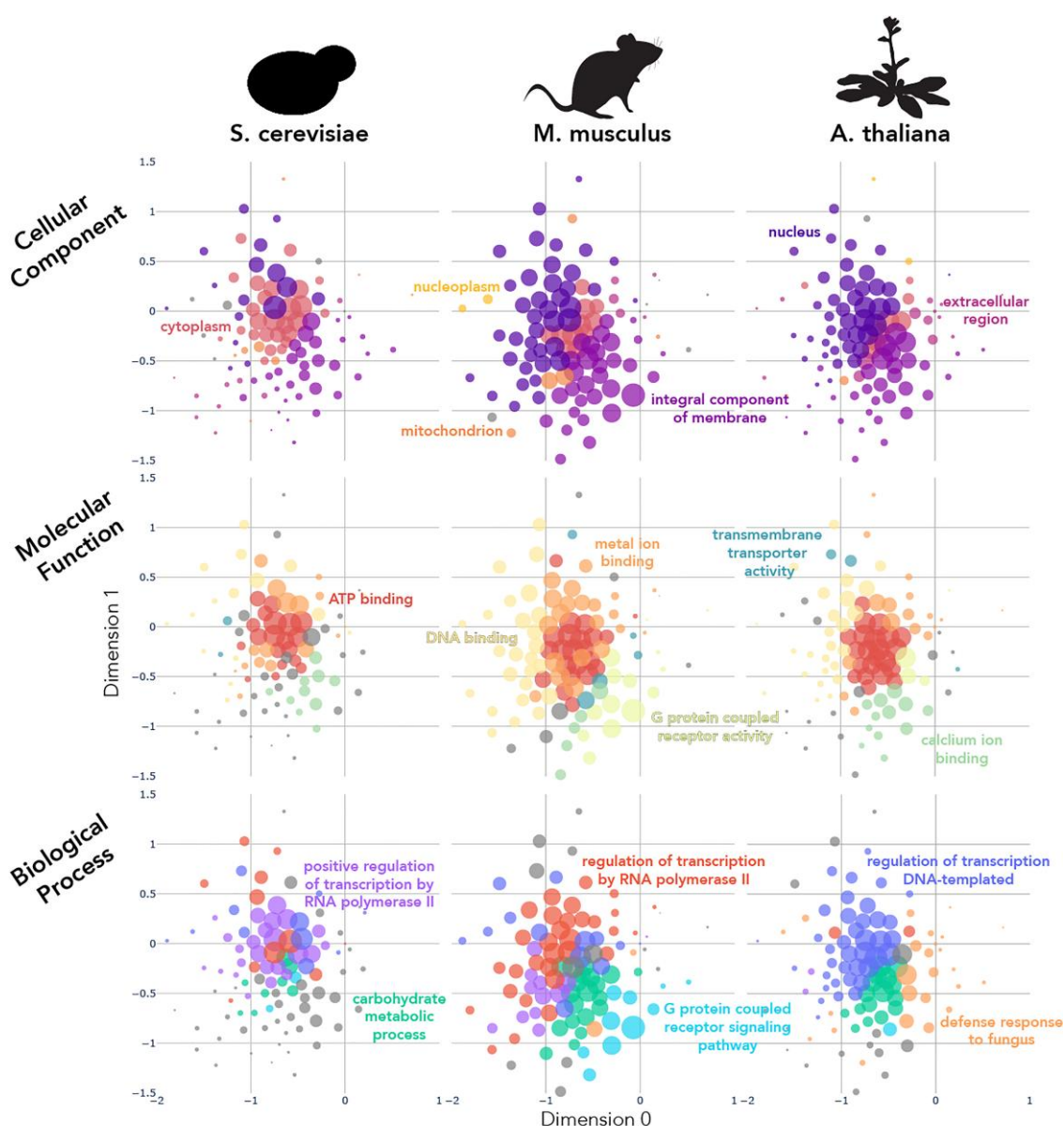


Fig. 3.— Visualizations in two-dimensional latent space. Circles represent clusters, size of the circle represents size of the cluster, and circles are annotated by most popular ontology for that cluster of the top six most common ontologies total.

protein sequence data with any degree of protein or taxon sampling. Clusters produced by DeepSeqProt from multiple-taxon data sets may also serve as useful precursors for tree-building and homology-based methods such as OrthoFinder (Emms and Kelly 2015, 2019), an area that warrants further investigation.

Use Cases and Caveats

The primary motivation behind DeepSeqProt was to develop a method for exploring protein function that was

not reference dependent. However, DeepSeqProt is not a replacement for experimental inquiry, as all machine learning models ultimately need a ground truth to validate their results against. The inclusion of high-confidence reference data from model organisms will not contribute to the performance of the model itself; however, they are still necessary in order to interpret results.

For the purposes of this paper, we focus on analyzing proteome-scale data, though note that DeepSeqProt can be trained and run on any protein data sets of sufficient size in any context. Proteins found to be differentially expressed,

for example, can be analyzed on a model trained on background proteins to observe whether up/downregulated proteins cluster together and where such cluster(s) are distributed in latent space. Similarly, although we discuss a very general model trained across eukaryotes here, researchers may train on any taxon or clade they prefer. Indeed, we may expect to see greater resolution and increased model performance on data sets tailored toward more exclusive clades.

Materials and Methods

To investigate the viability of unsupervised deep learning for protein identification, we wrote a VQ-VAE (van den Oord et al. 2017) program, called DeepSeqProt, whose general architecture is detailed below (fig. 1). DeepSeqProt operates by clustering proteins together on the basis of shared discrete embeddings learned during training. The distribution of these clusters in reduced dimensional space can then be used to infer their functional affinity. Our primary motivation was to develop a simple, but useful, deep learning framework for unaligned, unannotated, protein sequence data. DeepSeqProt is written in Python3 and uses the PyTorch (Paszke et al. 2019) machine learning library. More details are available at DeepSeqProt's GitHub repository (<https://github.com/KyleTDavid/DeepSeqProt>). DeepSeqProt can also be run interactively in the public Google Colaboratory Notebook available at <https://colab.research.google.com/drive/1FcOCECzhUg35PcXjb-LfdsSYDQoZurV3?usp=sharing>.

Data Preprocessing

Before applying neural networks to categorical amino acid data, sequences must first be converted into numeric form. DeepSeqProt accomplishes this through one-hot encoding, where each of the 20 canonical amino acids are translated to a unique binary array 20 elements long. Alanine, for example, is represented as (1, 0...0, 0), and valine is represented as (0, 0...0, 1). Any character not associated with one of the canonical amino acids (e.g., unknowns, ambiguity codes, and nonstandard amino acids) is represented by an array of all zeroes. One necessary restriction of deep learning is that most operations require input data to have the same shape, which presents a problem for amino acid sequence data which can be of variable length. Some methods overcome this limitation either by padding sequences with additional zeroes up to a standard length (Seo et al. 2018) (while discarding or truncating longer sequences) and/or by grouping sequences of the same/similar length together^{13,16}. However in the latter case, we worry that artificially ordering training data a priori may introduce unintended bias during training. We also experimented with zero-padding, but this caused the model to become fixated on sequence length, distributing clusters along an axis correlating with sequence size alone. DeepSeqProt

instead performs a linear interpolation as implemented in the SciPy library (Virtanen et al. 2020) from the one-hot encodings of each amino acid to an arbitrarily defined standard sequence length (fig. 1). In this way, each input sequence can be thought of as 20 one-dimensional arrays, or "channels," of a standardized arbitrary length, where each channel represents the relative contribution of a given amino acid along the length of the sequence. The default value of this length is 2,000, which is longer than 99.99% of sequences used in this analysis. However, we encourage users to modify this parameter to a value which makes most sense for their data set.

Encoder/Decoder

All VAEs contain an encoder and a decoder. Encoders and decoders are both neural networks; encoders are designed to compress input data into a reduced latent representation, whereas decoders are designed to reconstruct input data by sampling from the reduced latent space. Alternative dimensionality reduction techniques (t-SNE (Van der Maaten and Hinton 2008) and UMAP (McInnes et al. 2018)) were also tried but did not perform well. Similarly, many different encoder/decoder schemata were tested, with multiple convolutional layers and residual neural network architectures. DeepSeqProt uses a fully connected feedforward network (table 1). The depth and width of the network is provided by the user as a vector. For example, the default architecture of (2,000, 1,500, 1,000, 500, 1) will consist of four hidden layers. The first value in the vector represents the standard length to which all input sequences will be interpolated, and each successive value represents the "width" of each hidden layer in order. In our example, the first hidden layer is 1500 "neurons" wide and is fed by fully connected linear transformations from the encoded input sequence, which have all been interpolated to a standard length of 2,000. These values will then be passed through to the next hidden layer which is 1,000 neurons wide and so on. Note that each layer should have the same or fewer neurons to the one before it to accommodate the formulation behind autoencoders, which expect a progressive compression of input data. The decoder architecture is always the inverse of the encoder. To continue the example, the first hidden layer of the decoder will be a single neuron and ultimately terminate in a vector the same shape as the encoder input, as is necessary to estimate loss.

After traversing fully connected layers, but prior to vector quantization, DeepSeqProt also provides the option for a single convolutional layer (table 1). This layer allows data to be compressed further along the 20 amino acid channels, which is necessary to visualize the latent space in 2 or 3 dimensions. The default kernel size, stride, and dilation of this layer are all equal to one to preserve the data shape.

Naturally, if this option is selected, a complementary transposed convolutional layer, or “deconvolutional” layer, is automatically applied at the start of the decoder (table 1).

Vector Quantization

After passing through the encoder, data is in shape (B, C, W) , where B is the batch size, C is the number of channels (20, unless the optional convolution was applied), and W is the width of the final layer of the decoder. The first step of the vector quantization layer is to flatten data to N vectors of length D , where D is the number of dimensions in latent space. Recall that the motivation for the vector quantization step is to provide a discrete distribution in latent space in hopes that the model will use this distribution as a type of classifier for categorical input data. As we want each sequence to be associated with a single cluster, our encoded data must be flattened to shape $(N=B, D=C \times W)$. The number of possible clusters to the model is arbitrary and specified by the user. The default number is 1,000 though in development the model rarely used more than 100. If users wished to force the model to use a binary classifier, for example, they could use two clusters instead.

After encoded data have been flattened into the same dimensionality as the embedding vectors, distances are calculated from the encoded data to each of the embedding vectors. The closest vector is then reshaped back to (B, C, W) and passed to the decoder.

Loss and Learning

A noteworthy feature of VQ-VAEs compared with conventional VAEs is that the closest embedding lookup step has no gradient and therefore cannot be trained through backpropagation. VQ-VAEs can circumvent this by simply copying the gradient from decoder input to encoder output (van den Oord et al. 2017), similar to the straight-through estimator for stochastic neurons (Bengio et al. 2013). The loss function itself is made up of three components, reconstruction loss, codebook loss, and commitment loss. Reconstruction loss optimizes the encoder and decoder by encouraging the decoder to compute a faithful reconstruction of data x' from original input data x . Since reconstruction loss receives no gradient from embeddings, codebook loss is used to update embedding e closer to the encoder output $z_e(x)$. Finally, commitment loss prevents the latent space from growing arbitrarily large by encouraging the encoder to commit to an embedding and is modified by a weight hyperparameter β . During training, reconstruction loss is measured by the mean squared error of input data from reconstructed data, divided by the data variance. Likewise, codebook loss and commitment loss are measured by the mean squared error of the embeddings from the encoder output. The complete loss function can

therefore be represented as:

$$L = \log p(x' | z_q(x)) + \| \text{sg}[z_e(x)] - e \|_2^2 + \beta \| z_e(x) - \text{sg}[e] \|_2^2$$

where sg indicates the stop gradient operator to indicate that these values are not directly updated by the gradient during training. Prior to loss estimation, the embeddings themselves are further updated by exponential moving averages (EMA). EMA optimization was suggested (though not implemented) by the original authors of VQ-VAE, and we found that it performed better for our data set than the original method (van den Oord et al. 2017). Training stops automatically if the average loss of a sliding 1,000 update window fails to improve or if the maximum number of training updates set by the user is reached. DeepSeqProt uses the Adam optimization algorithm (Kingma and Ba 2014) for gradient descent and to update learning parameters.

Benchmarking

The aim of DeepSeqProt is to learn latent embeddings associated with specific groups of protein sequences. In this sense, DeepSeqProt can be thought of as a clustering tool, albeit with added information from the structure of latent space. We benchmarked DeepSeqProt alongside several clustering algorithms designed for unaligned protein sequences. CD-HIT (Li and Godzik 2006) was one of the first (Li et al. 2001) clustering algorithms designed for large databases and uses short word filtering followed by greedy incremental clustering by sequence similarity (Li and Godzik 2006; Fu et al. 2012). CD-HIT uses a clustering threshold parameter which controls the short word filtering step, for which we tested two values, 50% and 75%. MMseqs2 (Steinegger and Söding 2017) is a similar software suite designed to quickly and sensitively search and cluster large sequence data sets (Steinegger and Söding 2017). MMseqs2 also has a minimum sequence identity parameter similar to CD-HIT's clustering threshold for which we tested values 25%, 50%, and 75%. Finally, we performed Markov clustering on a similarity matrix generated from an all versus all DIAMOND (Buchfink et al. 2021) search. Each clustering algorithm was run on three proteomes, *Saccharomyces cerevisiae* (UP000002311), *M. musculus* (UP000000589), and *Arabidopsis thaliana* (UP000006548), from the UniProt protein database (Consortium 2018). We selected several summary statistics to measure the accuracy and precision of each method.

First, we wanted to see how well DeepSeqProt clusters represent entire protein families, as defined by UniProt. To achieve this, we calculated how many representatives from each family were captured within the same cluster. We call this metric “family completeness,” formally defined

as $\sum_{i=1}^I \frac{\text{Arg max } \{y_{i,1} \dots y_{i,n}\}}{T}$ where I is the total number of protein families, y is the number of representatives from protein family i across n clusters, and T is the total number of protein sequences in the data set. In other words, family completeness is the greatest fraction of sequences that are categorized together with other members of the same family, averaged across all families in the data set. We also calculated the AMI (Vinh et al. 2010) between protein families and clusters. Mutual information (Shannon 1948; Kreer 1957), or information gain, is a measure of mutual dependence between two random variables. AMI adds a correction to account for agreement due to random chance. Importantly, AMI is agnostic to the number of clusters and can be used to make fair comparisons across different methods. An AMI of 0 means that assigned labels (clusters) are totally incomplete, and an AMI of 1 means that labels are totally complete (each member of a family belongs to the same cluster) and homogeneous (each cluster only contains a single family).

Finally, we are interested in how well DeepSeqProt clusters together proteins of similar function. To this end, we performed gene ontology enrichment analysis (Klopfenstein et al. 2018) on each cluster produced by DeepSeqProt and our benchmark clustering algorithms (supplementary table S1, Supplementary Material online). From these, we calculate the fraction of proteins that possess at least one of the gene ontology terms found significant ($P < 0.01$ after Benjamini–Hochberg (Benjamini and Hochberg 1995) correction) for that protein's assigned cluster. This metric, referred to hereafter as "GO accuracy," is formally represented by the equation $\frac{\sum_{i=1}^J n(G)}{n(G)}$ where $n(G)$ is the number of proteins in a set G with at least one significant gene ontology term in a given cluster across J clusters. A GO accuracy of 0 means that no proteins have gene ontologies that are significantly enriched within their clusters, and a GO accuracy of 1 means that all of them are. Notably, each of these summary statistics has a lower bound at or approaching 0 and an upper bound of 1. In addition to information gained from which proteins are associated with which clusters, the embedding coordinates underlying each cluster may reflect local and global structure within functional space. To explore how deep learning may be used to visualize high dimensional protein function, we reran DeepSeqProt with an additional convolutional layer, restricting the embedding vectors two dimensions to better visualize the latent space.

Supplementary Material

Supplementary data are available at *Genome Biology and Evolution* online (<http://www.gbe.oxfordjournals.org/>).

Acknowledgments

The authors would like to thank all members of the Halanych lab for their valuable support. This work was supported by NSF grant OPP-1916661 and the NSF Graduate Research Fellowship (NSF-DGE-1937964).

Data availability

DeepSeqProt and all scripts used for benchmarking and validation are available at <https://github.com/KyleTDavid/DeepSeqProt>. This repository also contains pickled Python objects of both trained models discussed in this paper. An interactive version of DeepSeqProt is also available at <https://colab.research.google.com/drive/1FcOCECzhUg35PcXjb-LfdsSYDQoZurV3?usp=sharing>.

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Associate editor: Brian Golding